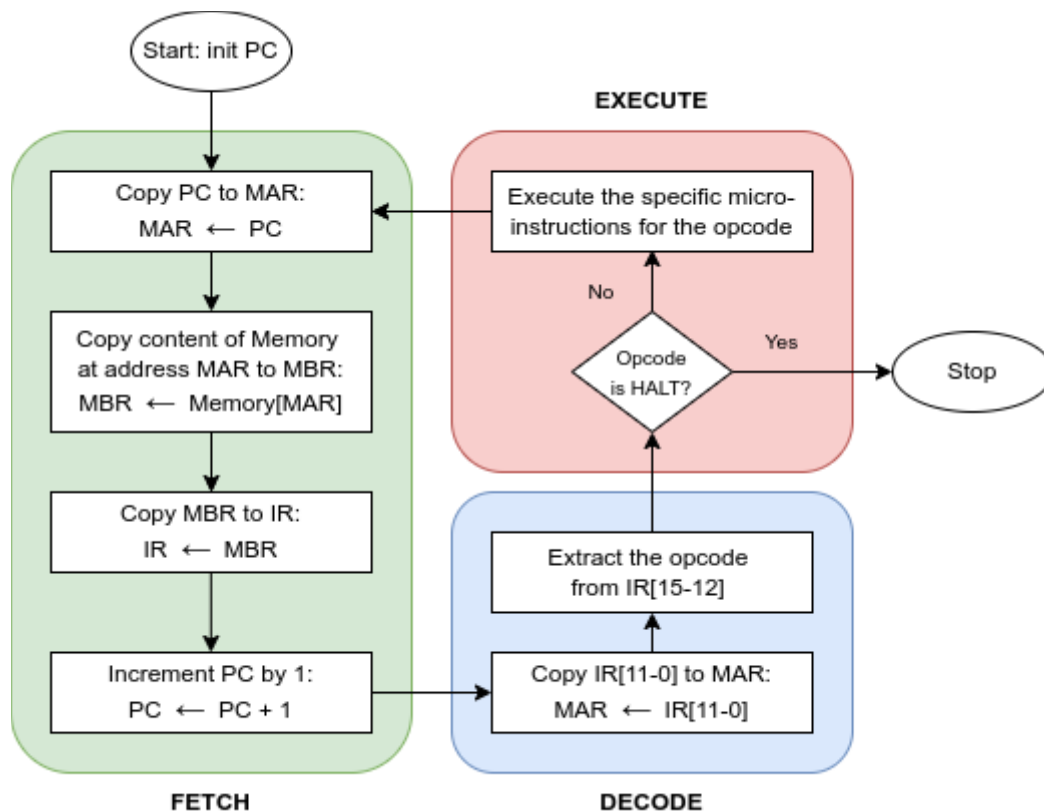


Fetch-Decode-Execute Cycle



Instruction Set

Opcode	Instruction	RTL of the Execute stage
1	Load X	$MBR \leftarrow Memory[MAR]; AC \leftarrow MBR$
2	Store X	$MBR \leftarrow AC; Memory[MAR] \leftarrow MBR$
3	Add X	$MBR \leftarrow Memory[MAR]; AC = AC + MBR$
4	Subt X	$MBR \leftarrow Memory[MAR]; AC = AC - MBR$
5	Input	$AC \leftarrow InReg$
6	Output	$OutReg \leftarrow AC$
7	Halt	Stop
8	Skipcond X	If $IR[11-10] == 00$ Then // Skipcond 000 If $AC < 0$ Then $PC \leftarrow PC + 1$; Else If $IR[11-10] == 01$ Then // Skipcond 400 If $AC == 0$ Then $PC \leftarrow PC + 1$; Else If $IR[11-10] == 10$ Then // Skipcond 800 If $AC > 0$ Then $PC \leftarrow PC + 1$;
9	Jump X	$PC \leftarrow IR[11-0]$
0	JnS X	$MBR \leftarrow PC; Memory[MAR] \leftarrow MBR; MBR \leftarrow X; AC \leftarrow 1; AC \leftarrow AC + MBR; PC \leftarrow AC$
A	Clear	$AC \leftarrow 0$
B	AddI X	$MBR \leftarrow Memory[MAR]; MAR \leftarrow MBR; MBR \leftarrow Memory[MAR]; AC \leftarrow AC + MBR$
C	JumpI X	$MBR \leftarrow Memory[MAR]; PC \leftarrow MBR$
D	LoadI X	$MBR \leftarrow Memory[MAR]; MAR \leftarrow MBR; MBR \leftarrow Memory[MAR]; AC \leftarrow MBR$
E	StoreI X	$MBR \leftarrow Memory[MAR]; MAR \leftarrow MBR; MBR \leftarrow AC; Memory[MAR] \leftarrow MBR$